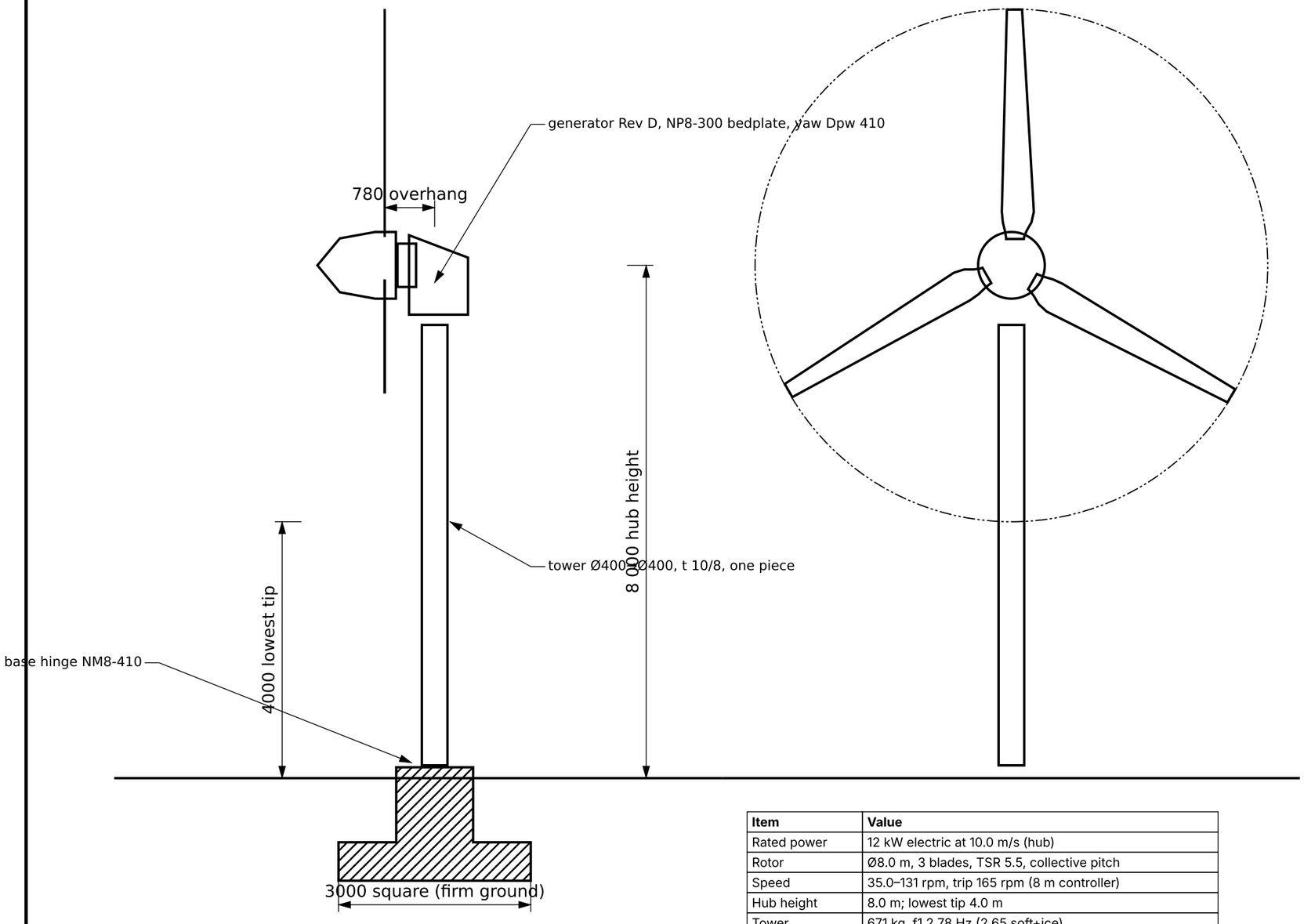
SIDE VIEW (wind from the left)



VIEW FROM UPWIND

AXONOMETRIC VIEW (not to scale)



Item	Value
Rated power	12 kW electric at 10.0 m/s (hub)
Rotor	Ø8.0 m, 3 blades, TSR 5.5, collective pitch
Speed	35.0–131 rpm, trip 165 rpm (8 m controller)
Hub height	8.0 m; lowest tip 4.0 m
Tower	671 kg, f1 2.78 Hz (2.65 soft+ice)
Exclusion zone	48–61 rpm (3P × f1)
Head mass	1362 kg above the tower flange (CAD)
Energy vs 18 m	−16 % highland … −27 % inland
Stopping	pitch to feather (supercap) + 3 spring calipers 14.4 kNm
Storm	feathered, idling or braked, yaw active

NOTES

1. Docs 35 and 37. Shared 12 kW parts keep their drawings (index NM8-001). Generator Rev D change sheet NM8-215.
2. Keep people and vehicles out of the rotor disc: fenced zone of radius 5 m round the tower base; lowest tip 4.0 m. Ice-fall zone 1.5 × (H + D) = 24 m when icing is possible.
3. Siting at 8 m: rotor bottom must be clear of the wake of buildings and shelter belts (≥ 20 × obstacle height upwind in the main wind directions); snow drift must stay below 2.5 m.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE M8 (ICELAND) · Ø8.0 m, COLLECTIVE PITCH · 8 m TOWER

GENERAL ARRANGEMENT · 12 kW M8 ON 8 m TOWER

DRAWING NO.	REV / DATE	SCALE (A3)
NM8-000	A · 2026-11-06	1:75
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
see BOM NM8-002/003	≈ 1362 kg head + 671 kg tower	Claude / not checked

Drawing	Title	Source
NM8-000	GENERAL ARRANGEMENT · 12 kW M8 ON 8 m TOWER	M8 rev A
NM8-001	DRAWING INDEX	M8 rev A
NM8-002	BILL OF MATERIALS (1/2)	M8 rev A
NM8-003	BILL OF MATERIALS (2/2)	M8 rev A
NM8-010	GENERAL ARRANGEMENT · NACELLE AND DRIVETRAIN	M8 rev A
NM8-020	ROTOR ASSEMBLY · HUB, PITCH SYSTEM, BLADES, SPINNER	M8 rev A
NM8-100	BLADE Ø8.0 m · PLANFORM, TWIST AND ROOT	M8 rev A
NM8-105	BLADE ROOT INSERT M16 (BONDED)	M8 rev A
NM8-110	BLADE MOULD SET (PLUG, TWO HALVES, FRAME)	M8 rev A
NM8-200	HUB WELDMENT (COLLECTIVE PITCH)	M8 rev A
NM8-210	SPINNER (GRP) WITH BLADE BOOTS	M8 rev A
NM8-215	GENERATOR REV D · CHANGE SHEET (STACK 74 → 104 mm)	M8 rev A
NM8-250	COLLECTIVE PITCH · BEARINGS, CRANKS, LINKS, YOKE, CYLINDER	M8 rev A
NP8-300	BEDPLATE WELDMENT 12 kW NP8 (LIGHT)	M8 rev A
NM8-311	ROTOR BRAKE · THREE CALIPERS AND THIRD CALIPER BRACKET	M8 rev A
NM8-911	YAW SYSTEM LAYOUT · SLEWING RING Dpw 410	M8 rev A
NM8-400	TOWER 8 m · ONE-PIECE TILT-UP MONOPOLE	M8 rev A
NM8-410	BASE HINGE, GIN POLE INTERFACE AND ERECTION	M8 rev A
NM8-500	FOUNDATION · SPREAD FOOTING (8 m TOWER, M8)	M8 rev A
NM8-900	FASTENER AND TORQUE SCHEDULE	M8 rev A
NM8-910	BOUGHT PARTS SPECIFICATION (ENVELOPES AND INTERFACES)	M8 rev A
NP1-101/102	Front spacer, rear seal sleeve (Rev B)	12 kW P1 rev B
NP1-202	Stator front clamp ring	12 kW P1 rev B
NP1-211	Bearing seal covers front / rear	12 kW P1 rev B
NP1-212C	Drum rear cover with 24 fan blades	Rev C (doc 23)
NP1-230	Brake disc 16 mm Ø880 / Ø652	12 kW P1 rev B
NP1-310	Brake caliper brackets 3 and 9 o'clock	12 kW P1 rev B
NP1-320/321	Rotor lock pin and holder	12 kW P1 rev B
NP1-420	Yaw drive brackets (radius +30)	12 kW doc 25
NY-400…460	Adapter, pad units, lock, encoder, cable hanger (radii +25, see NM8-410)	12 kW doc 25
NM8-700/710/720/730	Nacelle cover, frame, met mast, junction box (S6 parts, cover front 15 mm longer	S6 rev B
NM8-420	Gin pole and tilt-up rigging (S6 tool, line pull see NM8-400)	S6 rev B
NM8-600/620/800	Resistor box, base cabinet, cabling routes (S6)	S6 rev B

NOTES

1. Rev A (2026-11-06): first issue for the 12 kW M8 on the 8 m tower (doc 37). Every new or changed part has an axonometric view from the 3D model (cad/turbine_m8/model8.py).
2. Reused drawings are used unchanged except where this index says otherwise. Their PDFs are in the earlier sets (cad/nacelle_p1, cad/yaw, cad/turbine_6kw).
3. All sheets NOT CHECKED: prototype release only after review, the open checks of doc 37 §9 and the supplier drawings of the bought parts.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE M8 (ICELAND) · Ø8.0 m, COLLECTIVE PITCH · 8 m TOWER

DRAWING INDEX (NM8 SET REV A + REUSED DRAWINGS)

DRAWING NO.	REV / DATE	SCALE (A3)
NM8-001	A · 2026-11-06	–
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
–	–	Claude / not checked

Pos	Part no.	Description	Qty	Make/buy	Material / spec	kg each
		A ROTOR				
1	NM8-100	Blade Ø8.0 m, glass/epoxy, carbon UD caps (B1–B3)	3	make	E-glass/epoxy, carbon UD caps, PVC core	34.1
2	NM8-105	Root insert M16 (12 per blade, bonded)	36	make	C45E+QT zinc-flake coated	0.8
3	NM8-200	Hub weldment: shell Ø530 × 10, 3 necks Ø360, drum flange	1	make	S355NL, welded, stress relieved, machined	179.8
4	NM8-210	Spinner (GRP 4 mm) with blade boots	1	make	glass/polyester, gelcoat RAL 9010	27.6
5	NM8-250	Pitch bearings Dpw 300 four-point (3, bought)	1	buy	bought: 42CrMo4 rings	76.2
6	NM8-260	Pitch cranks r140 with pins (3)	1	make	42CrMo4+QT / S355 tube	30.0
7	NM8-261	Pitch links 220 mm with spherical eyes (3)	1	make	S355 / bought rod ends	4.5
8	NM8-262	Yoke + 3 guide rods	1	make	S355 plate / C45 chromed rods	8.0
9	NM8-270	Electric pitch cylinder ~28.5 kN rated, 204 mm stroke + motor	1	buy	bought	14.0
10	NM8-280	Supercap module + pitch drive/controller (bought)	1	buy	bought	6.5
11	NM8-290	Nose frame (actuator + electronics carrier)	1	make	S355 rings + tube struts	6.0
		B DRIVETRAIN AND GENERATOR REV D				
12	NM8-311	Third caliper bracket (135°)	1	make	S355J2+N 25 plate, welded	7.0
13	NM8-913	Tapered roller bearing 32028 X (front + rear)	2	buy	bought 32028 X	4.0
14	NM8-914	Overspeed sensors M18 inductive (2, bought)	1	buy	bought M18 PNP	0.4
15	NM8-915	Spring-applied brake caliper 4.8 kNm (3 on the disc)	3	buy	bought, spring-applied	9.0
16	NP1-100D	Spindle Rev D (30 mm longer between the rear seat and the flan	1	make	42CrMo4+QT forged	59.2
17	NP1-101	Front spacer / seal counterface Ø180 (Rev B)	1	make	42CrMo4+QT, induction hardened	1.4
18	NP1-102	Rear seal sleeve Ø195 on the Ø155 shoulder (Rev B)	1	make	42CrMo4+QT, induction hardened	1.3
19	NP1-200D	Stator carrier Rev D, 45 dovetail grooves, rim 104 mm	1	make	S355J2+N	45.4
20	NP1-201D	Stator Rev D: 45 tooth-segment stacks, 104 mm (7 magnet rows)	1	make	M270-35A 0.35 mm, self-bonding	71.0
21	NP1-202	Stator front clamp ring	1	make	1.4404 (non-magnetic)	2.2
22	NP1-203D	Coils on segments, 45 × 28 turns, 104 mm stack	1	make	Cu enamelled Ø1.55 class 200, NKN 0.30	20.0
23	NP1-210D	Rotor drum weldment Rev D (shell +30; Rev C fins and ribs, see	1	make	S355J2+N	95.5
24	NP1-211-F	Front bearing seal cover	1	make	S355J2	1.5
25	NP1-211-R	Rear bearing seal cover, bore Ø198 (Rev B)	1	make	S355J2	1.4
26	NP1-212C	Drum rear cover with 24 fan blades (Rev C; blades not modelled	1	make	EN AW-5083-H111	7.2
27	NP1-220F	Magnets 280 × N45SH 30×15×12 in 7 rows + comb + sleeve	1	make	NdFeB N45SH	15.0
28	NP1-230	Brake disc Ø880/Ø652 with 12 lock holes (Rev B)	1	make	C45E+N 16 mm	33.6
29	NP1-310	Rotor brake caliper bracket (left, Rev B)	1	make	S355J2+N 25 plate	18.1
30	NP1-320	Rotor lock pin Ø20 (engaged, top)	1	make	42CrMo4+QT	0.5
31	NP1-321	Lock pin holder with 2 overspeed sensor seats	1	make	S355J2	24.5
		C NACELLE				
32	NM8-700-1	Nacelle cover, lower tray (GRP 5 mm)	1	make	glass/polyester, gelcoat RAL 9010	15.7
33	NM8-700-2	Nacelle cover, upper lid (GRP 5 mm)	1	make	glass/polyester, gelcoat RAL 9010	17.7
34	NM8-710	Cover support arms + rear rail	1	make	S235 galv.	8.3
35	NM8-720	Met mast: anemometer, vane, air terminal	1	make	1.4404 tube 42.4 × 2 + bought sensors	4.5
36	NP8-300	Bedplate weldment 12 kW NP8 (plates 20 / 30 / 10)	1	make	S355J2+N plates, stress relieved, machined	124.3
		D YAW SYSTEM				
37	NM8-911	Yaw slewing ring Dpw 410, z165 m4	1	buy	bought: 42CrMo4, induction hardened teeth	66.0
38	NM8-912	Yaw gearmotor 0.25 kW, output ≥ 2.3 kNm static, motor brake	2	buy	bought	19.0
39	NP1-420	Yaw drive bracket	2	make	S355J2+N	7.7
40	NY-400	Tower-top adapter with friction track	1	make	S355J2+N weldment	118.3
41	NY-410	Friction pad unit	2	make	S355 / DIN 2093 springs / organic pads	14.0
42	NY-430	Yaw lock (screw block)	1	make	42CrMo4 / S355	6.5
43	NY-440	Yaw encoder + cam limit switch unit	1	make	S355 + absolute encoder + cam switch	6.0
44	NY-460	Cable hanger tube + loop cleat	1	make	S355 tube 168.3 × 4.5	10.0
		E ELECTRICAL				
45	NM8-600	Braking / dump resistor box (vented)	1	make	1.4301 box + steel-grid resistors	18.0
46	NM8-620	Tower base cabinet IP55 800 × 600 × 300 on stand	1	make	stand S235 galv.; bought enclosure	95.0
47	NM8-730	Nacelle junction box IP66 (400 × 340 × 150)	1	make	bought: 1.4301 enclosure	9.0

NOTES

1. Masses from the CAD model (cad/turbine_m8/model8.py) for made parts; supplier data or the doc 35 budget for bought and pitch parts. Full BOM: bom8.csv.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE M8 (ICELAND) · Ø8.0 m, COLLECTIVE PITCH · 8 m TOWER

BILL OF MATERIALS (1/2)

DRAWING NO.	REV / DATE	SCALE (A3)
NM8-002	A · 2026-11-06	–
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
see lines	see lines	Claude / not checked

Pos	Part no.	Description	Qty	Make/buy	Material / spec	kg each
48	NM8-800	Drop cable + base duct (cabling route)	1	make	H07RN-F 4G10 + 2 × LiYCY signal; PE duct Ø11	12.0
		F TOWER				
49	NM8-400	Tower 8 m, Ø400 → Ø400, t 10/8, one piece	1	make	S355J2+N, hot-dip galvanised	699.8
50	NM8-401	Hand-hole cover 200 × 350 × 6 with gasket	1	make	S355 galv.	3.3
51	NM8-410	Base hinge brackets + hinge pin Ø50	1	make	S355J2+N galv.; pin 42CrMo4+QT	37.4
		G FOUNDATION				
52	NM8-500	Spread footing (firm ground) with pedestal	1	make	C30/37 XF4 air-entrained, B500C	14880.0
53	NM8-510	Anchor cage 16 × M24 + 4 hinge anchors	1	make	M24 8.8 HDG threaded rod, S355 template ring	161.7
		H TOOLS AND MOULDS				
54	NM8-110-1	Blade mould, lower (pressure side) half on frame	1	make	GRP tooling laminate 12 mm, vinylester tooli	118.1
55	NM8-110-2	Blade mould, upper (suction side) half	1	make	GRP tooling laminate 12 mm	118.1
56	NM8-110-3	Mould frame (RHS 80 × 80 × 4)	1	make	S235 RHS 80×80×4	178.6
57	NM8-420	Gin pole 3.0 m (erection tool)	1	make	S355 tube 168.3 × 6.3	94.0
		J FASTENERS, CABLES, SITE MATERIALS				
58	NM8-900-01	Bolt M16 × 90 10.9 zinc-flake + washer (blade roots into the p	36	buy	10.9 zinc-flake	0.20
59	NM8-900-02	Bolt M16 × 70 10.9 (hub to drum)	12	buy	10.9 zinc-flake	0.17
60	NM8-900-03	Bolt M16 × 80 10.9 (pitch bearing outer rings to hub necks, 20	60	buy	10.9 zinc-flake	0.18
61	NM8-900-04	Bolt M20 × 90 10.9 (spindle to bedplate)	12	buy	10.9 zinc-flake	0.32
62	NM8-900-05	HV set M16 × 110 10.9 (yaw inner ring, PCD 350)	16	buy	EN 14399-4	0.26
63	NM8-900-06	Bolt M16 × 70 10.9 (yaw outer ring PCD 490, pads, calipers, br	40	buy	10.9 zinc-flake	0.17
64	NM8-900-07	HV set M20 × 90 10.9 (adapter to tower)	16	buy	EN 14399-4	0.40
65	NM8-900-08	Nut M24 8.8 HDG + plate washer (base 16 × 3, hinge 8)	56	buy	8.8 HDG	0.15
66	NM8-900-09	Small fasteners kit M6–M12 A4 (cover, spinner, mast, nose fram	1	buy	A4-70	3.00
67	NM8-800-W1	Cable H07RN-F 4G10 (generator → box)	3	buy	m	0.60
68	NM8-800-W2	Cable H07RN-F 4G10 (box → loop → hand hole)	12	buy	m	0.60
69	NM8-800-W3	Cable NYY-J 4 × 10 (hand hole → cabinet)	4	buy	m	0.50
70	NM8-800-W4	Cable ÖLFLEX 4G1.5 shielded (3 brake calipers)	20	buy	m	0.10
71	NM8-800-W5	Cable LiYCY 12 × 0.5 (sensors)	16	buy	m	0.12
72	NM8-800-W6	Cable ÖLFLEX 7G1.5 (yaw motors)	16	buy	m	0.15
73	NM8-800-W7	Slip ring 6 power + 6 signal (pitch supply and bus, in the nos	1	buy		1.50
74	NM8-800-W8	Earthing conductor Cu 50 mm² + ring earth + 2 rods 3 m	1	buy	Cu	18.00
75	NM8-500-M1	Concrete C30/37 XF4 air-entrained (firm ground)	6.2 m³	buy	ready-mix	
76	NM8-500-M2	Rebar B500C (mesh top + bottom, pedestal cage)	1	buy	≈ 460 kg	460.00
77	NM8-500-M3	Non-shrink grout ≥ 40 MPa, 30 mm under the base flange	70 l	buy		
78	NM8-600-K1	Contactor K1 3-pole NC (mirror contacts) + safety relay PL d	1	buy		2.00
79	NM8-620-PE	Converter 12 kW, dump chopper, AC module, 48 V module, control	1	make	in-house electronics	60.00

NOTES

1. Masses from the CAD model (cad/turbine_m8/model8.py) for made parts; supplier data or the doc 35 budget for bought and pitch parts. Full BOM: bom8.csv.

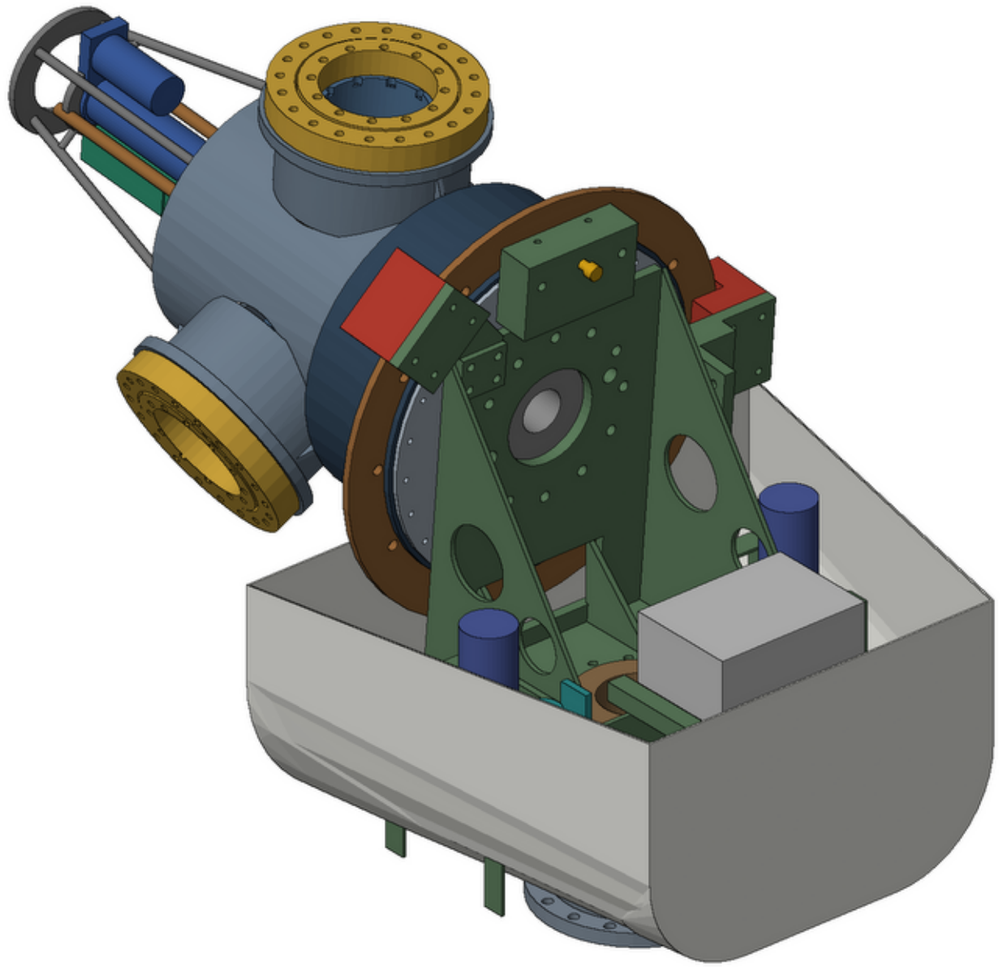
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE M8 (ICELAND) · Ø8.0 m, COLLECTIVE PITCH · 8 m TOWER

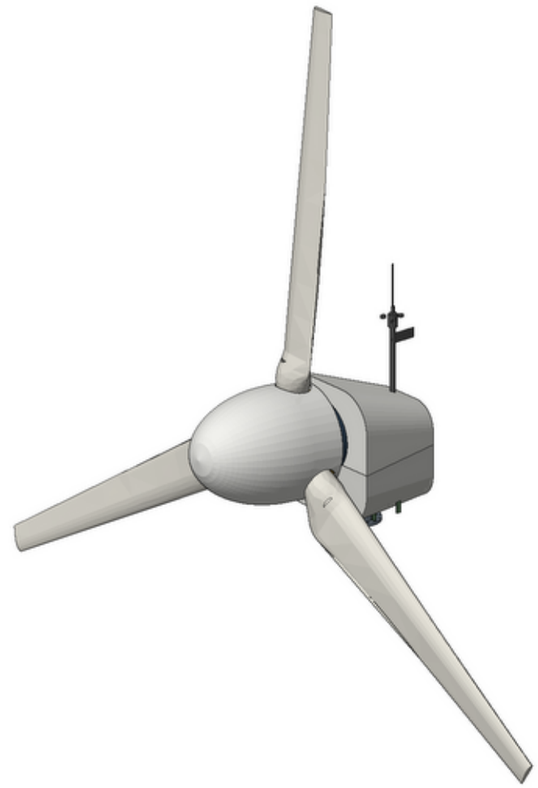
BILL OF MATERIALS (2/2)

DRAWING NO.	REV / DATE	SCALE (A3)
NM8-003	A · 2026-11-06	–
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
see lines	see lines	Claude / not checked

AXONOMETRIC VIEW (not to scale)



AXONOMETRIC VIEW (not to scale)



Interface	Value
Rotor plane to yaw axis	780 mm (doc 35 overhang; 750 + 30 Rev D)
Rotor centre above yaw bearing top	570 mm; shaft tilt 5°
Generator Rev D	stack 104 mm, 7 rows, 280 blocks, 28 turns
Stretch plane	spindle x 410: parts ahead move 30 mm forward
Bearings	2 × 32028 X, span 240 (unchanged)
Brake	3 spring calipers at 0°, 135°, 180° (lock pin at 90°)
Bedplate	NP8-300, 136 kg (plate model), util 0.68
Yaw	ring Dpw 410 z165 m4, 2 gearmotors, 2 pad units
Hub to drum	12 × M16 10.9 PCD 400, Ø360 spigot
Spindle to bedplate	12 × M20 10.9 PCD 350, Ø200 spigot

NOTES

1. A03: nacelle with the upper cover lid, met mast, spinner and blades removed, seen from the rear left. A02: complete head.
2. Whole-turbine interference check (cad/turbine_m8/clash8.py): no clashes left except intended fits (spigots, pins in eyes, bolts in tapped holes, embedded anchors, gear mesh contact) - list in doc 37 §6.
3. The hub, drum, stator and spindle front move 30 mm forward together, so all front interfaces (hub spigot, bearing seats, seal covers) are as drawn on the Rev B/C sheets; only the lengths in NM8-215 change.

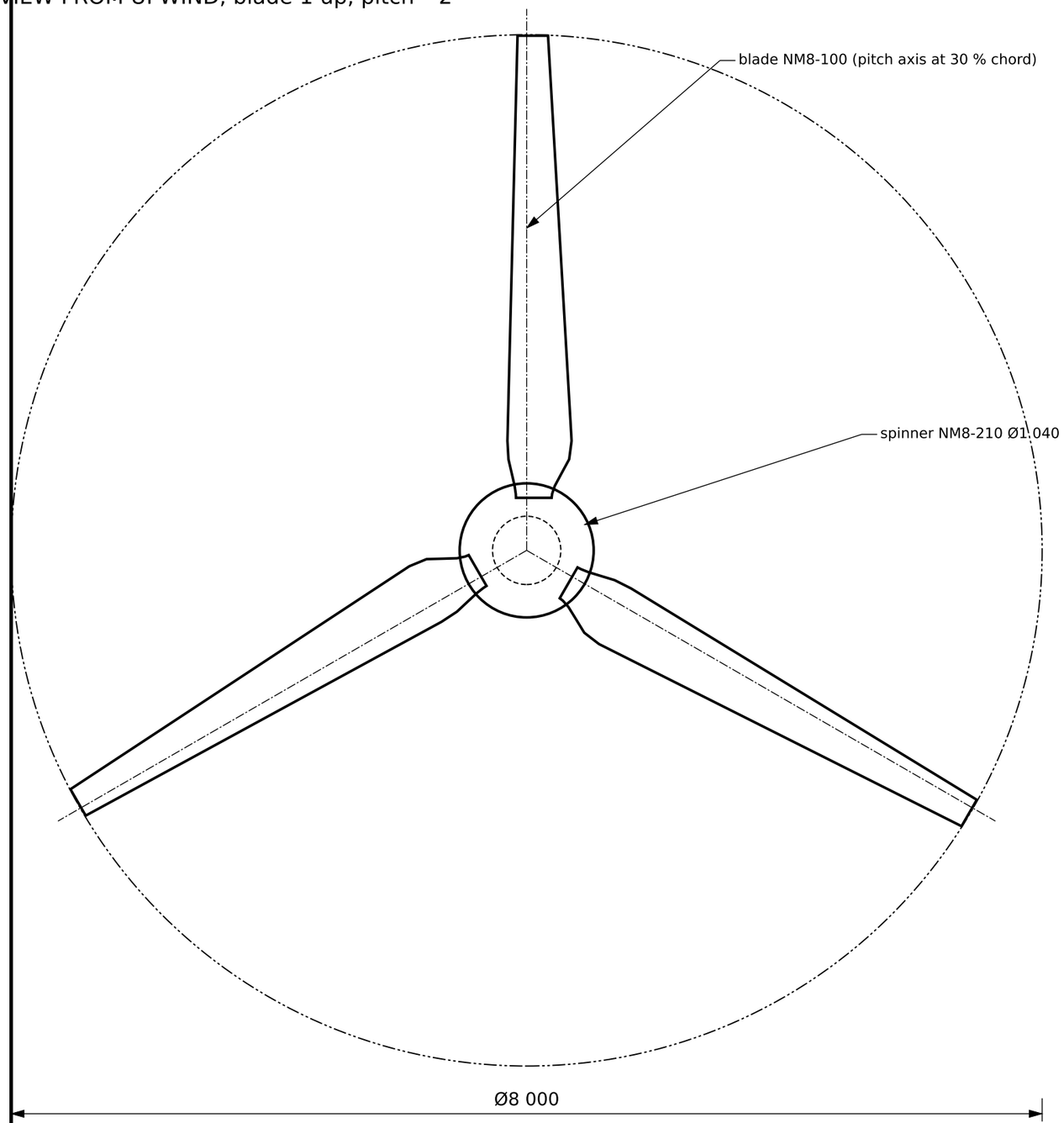
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE M8 (ICELAND) · Ø8.0 m, COLLECTIVE PITCH · 8 m TOWER

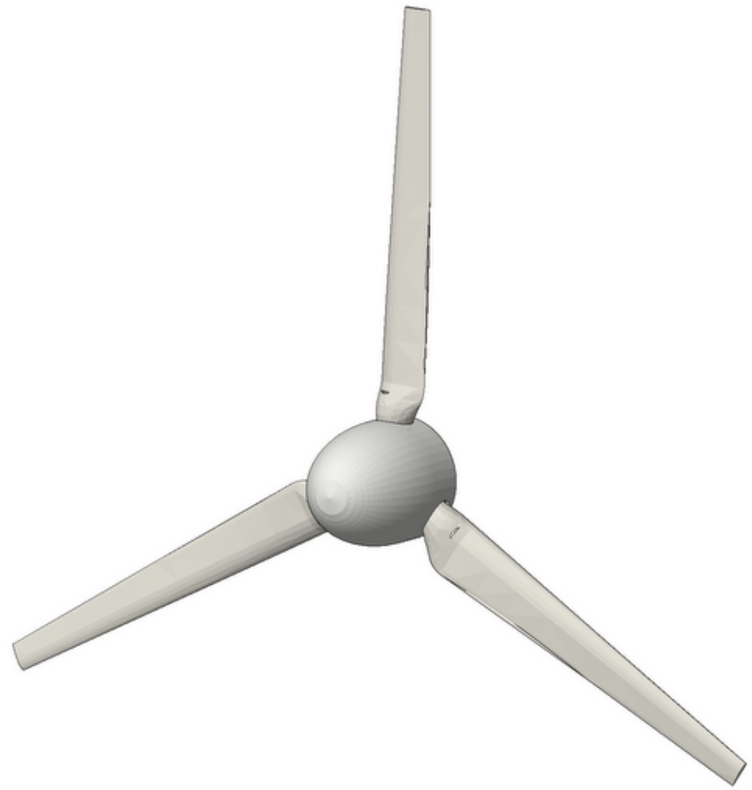
GENERAL ARRANGEMENT · NACELLE AND DRIVETRAIN

DRAWING NO.	REV / DATE	SCALE (A3)
NM8-010	A · 2026-11-06	–
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
see BOM	≈ 1362 kg above the tower flange	Claude / not checked

VIEW FROM UPWIND, blade 1 up, pitch −2°



AXONOMETRIC VIEW (not to scale)



Item	Value
Blades	3 × 34.1 kg, carbon UD caps (D3)
Blade root	Ø280, 12 × M16 inserts PCD 245
Pitch bearings	4-point Dpw 300, balls 30, Z 28
Pitch drive	one electric cylinder, crank r140, collective
Hub	S355NL weldment 180 kg (budget)
Rotor inertia	≈ 385 kg m²
Pitch range	−2° (fine) … 92° (feather)

NOTES

1. Assembly order (NM8-020 = build book steps 8–10): hub pre-assembled with bearings, cranks, links, yoke, cylinder and nose frame on the bench; mounted on the drum; blades fitted with the cylinder at the −2° end stop; spinner last.
2. Static-moment matched blade set: moments within 0.5 % (balance by root ballast pocket); mark each blade with its hub position.
3. Bolt torques NM8-900. Blade root bolts M16 10.9: 280 Nm, re-torque after 50 h.

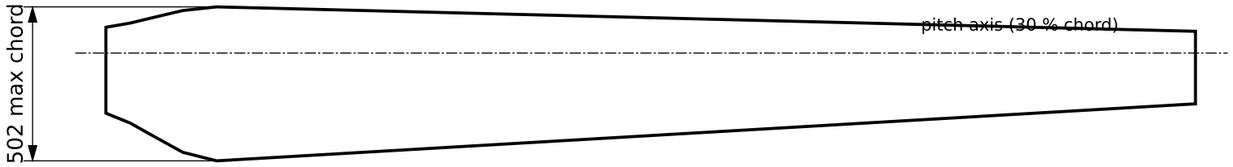
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE M8 (ICELAND) · Ø8.0 m, COLLECTIVE PITCH · 8 m TOWER

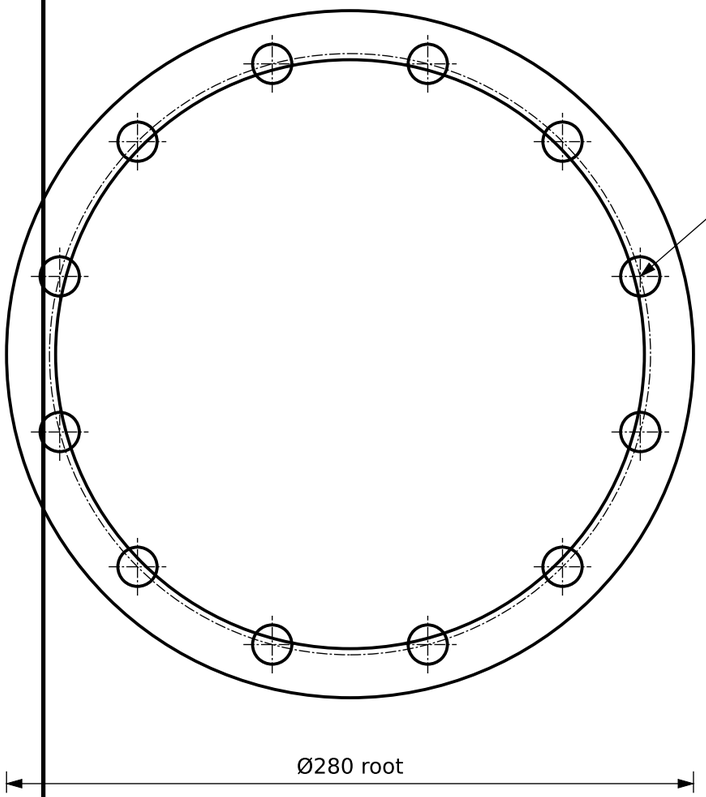
ROTOR ASSEMBLY · HUB, PITCH SYSTEM, BLADES, SPINNER

DRAWING NO.	REV / DATE	SCALE (A3)
NM8-020	A · 2026-11-06	1:40
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
see BOM	≈ 455 kg	Claude / not checked

PLANFORM (suction side up, leading edge up)



FACE



AXONOMETRIC VIEW (not to scale)



r mm	chord mm	twist °	t/c
450	280	20.0	0.91
600	389	20.0	0.64
800	501	20.0	0.30
1000	486	18.4	0.29
1400	452	12.6	0.26
1800	419	9.0	0.24
2200	386	6.5	0.22
2600	352	4.7	0.20
3000	319	3.4	0.19
3400	286	2.3	0.18
3800	253	1.5	0.18
4000	236	1.2	0.18

NOTES

1. Twist relative to the pitch reference at −2° setting (blade at its fine end stop). Aerofoil family as doc 10, thickness scaled to the 4.0 m span (structure_m8.py).
2. Laminate: doc 10 construction scaled; carbon UD caps sized by buckling. Mass 34.1 kg; flap frequency ≈ 18.8 Hz at rated (3P 6.6 Hz).
3. Root: 12 bonded inserts M16 on PCD 245 (bolt force 64 kN design, F_t,Rd 113 kN). Root face machined flat 0.1 after cure, square to the pitch axis 0.1°.
4. First-article static test SLM H × 1.35 flap, edge test, insert pull-out (3 per batch). Balance set to 0.5 % static moment.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

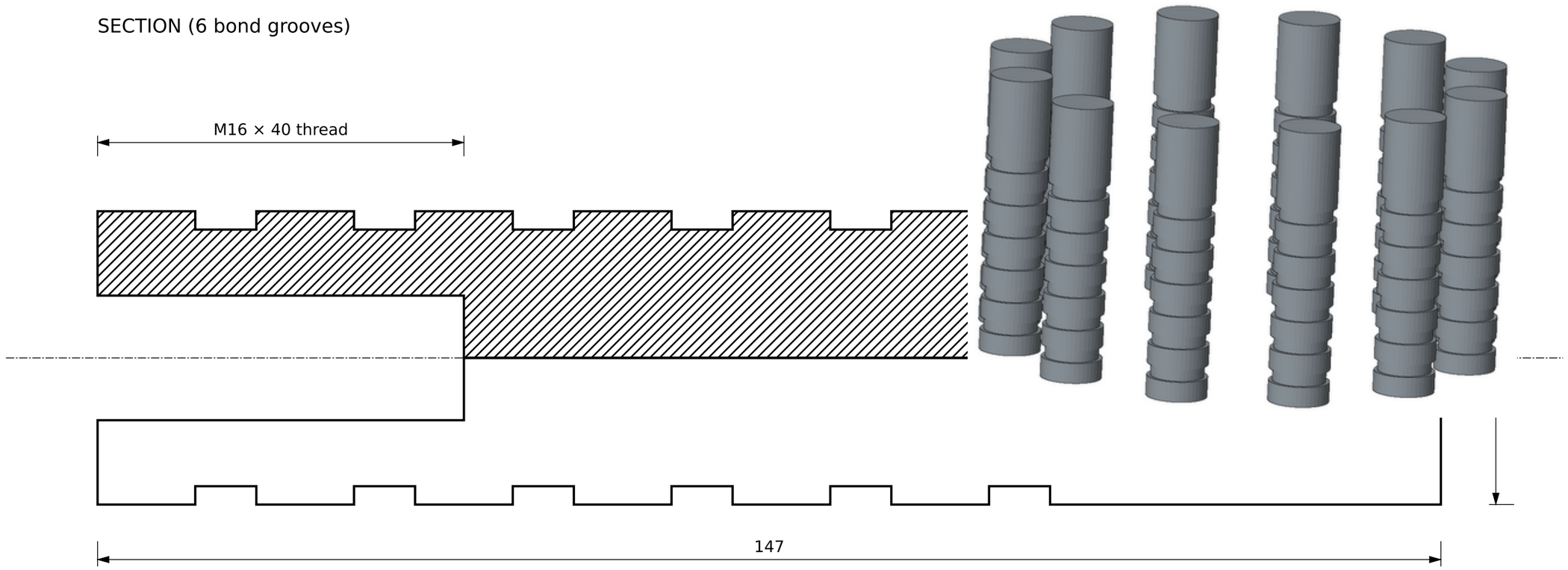
12 kW WIND TURBINE M8 (ICELAND) · Ø8.0 m, COLLECTIVE PITCH · 8 m TOWER

BLADE Ø8.0 m · PLANFORM, TWIST AND ROOT

DRAWING NO.	REV / DATE	SCALE (A3)
NM8-100	A · 2026-11-06	1:20
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
E-glass/epoxy infused; carbon UD caps; PVC core 34.1 kg		Claude / not checked

AXONOMETRIC VIEW (not to scale)

SECTION (6 bond grooves)



NOTES

1. The doc 34 M12 insert scaled 4/3: Ø32 × 147, internal M16-6H 40 deep, 6 grooves 2 deep for the bond.
2. Bonded into the root laminate with the NM8-115 fixture (12 spigots on PCD 245, as NS6-115 with the M8 pattern).
3. Pull-out test 3 per batch to 1.5 × 64 kN; thread gauge every insert.

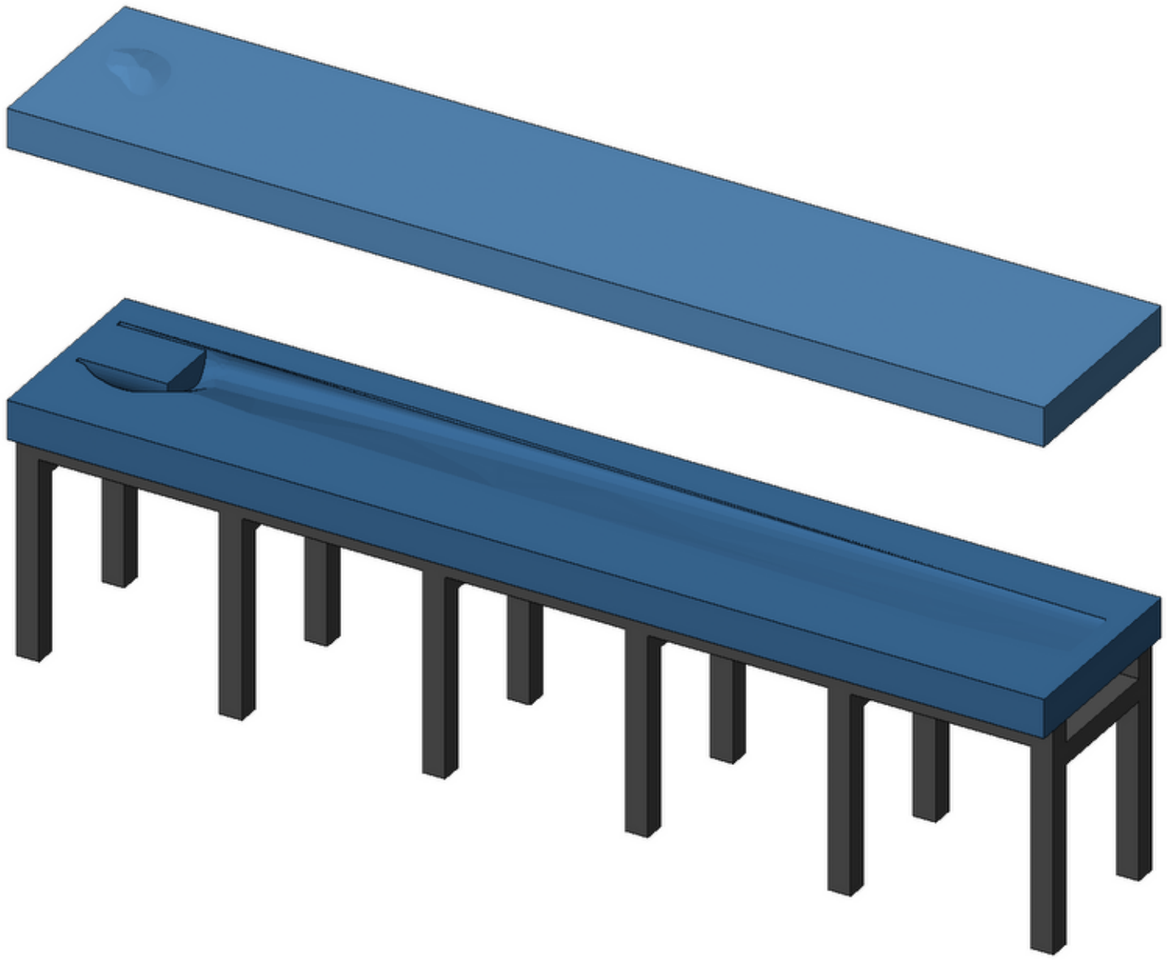
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE M8 (ICELAND) · Ø8.0 m, COLLECTIVE PITCH · 8 m TOWER

BLADE ROOT INSERT M16 (BONDED)

DRAWING NO.	REV / DATE	SCALE (A3)
NM8-105	A · 2026-11-06	2:1
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
C45E+QT, zinc-flake coated (Geomet 500)	0.78 kg	Claude / not checked

AXONOMETRIC VIEW (not to scale)



NOTES

1. Same construction as NS6-110 (doc 34), lengthened for the 3.56 m M8 blade: CNC-machined plug from the NM8-100 surfaces, GRP halves 12 mm with vinylester tooling gelcoat, steel frame RHS 80 × 80 × 4.
2. Parting line on the leading/trailing edge; root end with the NM8-115 insert fixture flange; 6 locating cones.
3. Mould acceptance: blade profile within 0.5 mm of the plug at 10 stations; twist within 0.2°.

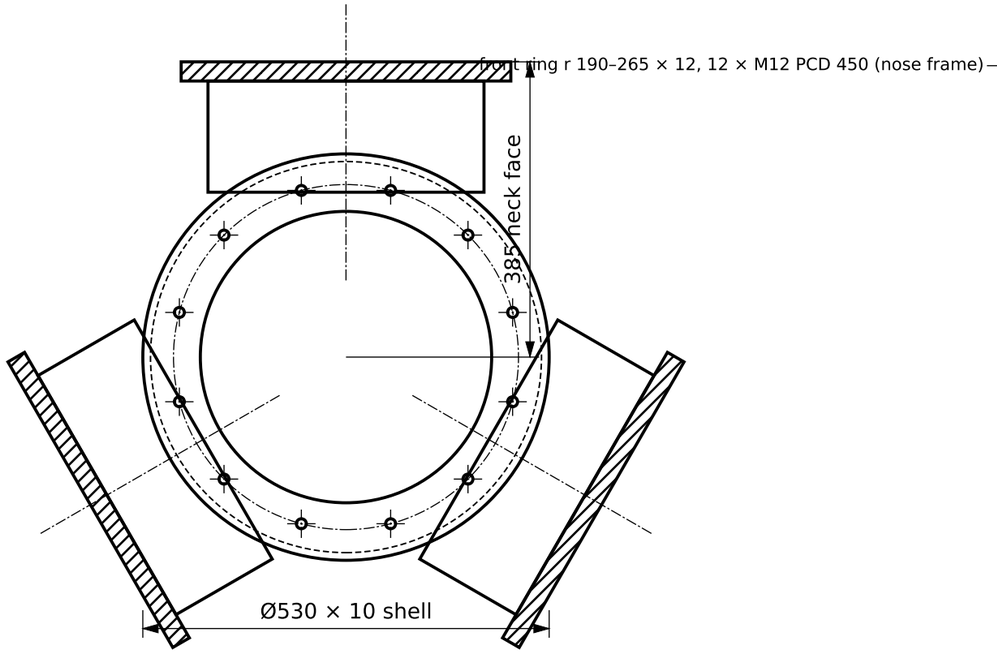
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE M8 (ICELAND) · Ø8.0 m, COLLECTIVE PITCH · 8 m TOWER

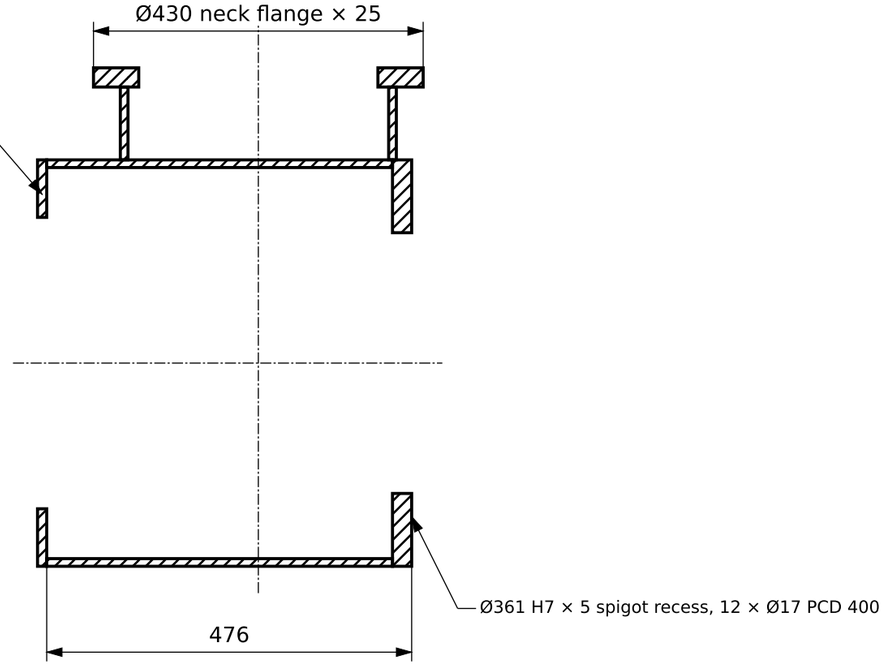
BLADE MOULD SET (PLUG, TWO HALVES, FRAME)

DRAWING NO.	REV / DATE	SCALE (A3)
NM8-110	A · 2026-11-06	–
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
GRP tooling laminate on S235 RHS frame	≈ 2 × 118 kg + frame 179 kg	Claude / not checked

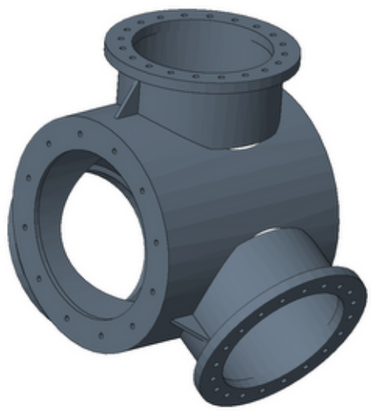
VIEW FROM UPWIND (spinner and nose frame removed)



SECTION THROUGH BLADE 1 AXIS



AXONOMETRIC VIEW (not to scale)



NOTES

1. Necks Ø360 × 10 on the Ø530 × 10 shell: parked SLM stress 117 MPa (allow 323); fatigue 31.5 MPa (allow 42.1).
2. Neck flanges Ø430 × 25 carry the pitch bearing outer rings: 20 × M16 tapped on PCD 360; faces machined after stress relief, flat 0.05, square to the neck axis 0.05.
3. Rear gussets stop on the drum flange ring (x 200) so they clear the drum web; front gussets 68 long. Full penetration welds ISO 5817 B, 100 % MT, UT on neck welds.
4. Drum joint as Rev C: Ø360 spigot, 12 × M16 10.9 PCD 400, 280 Nm. FE check of the neck-to-shell weld before release (doc 37 §9).

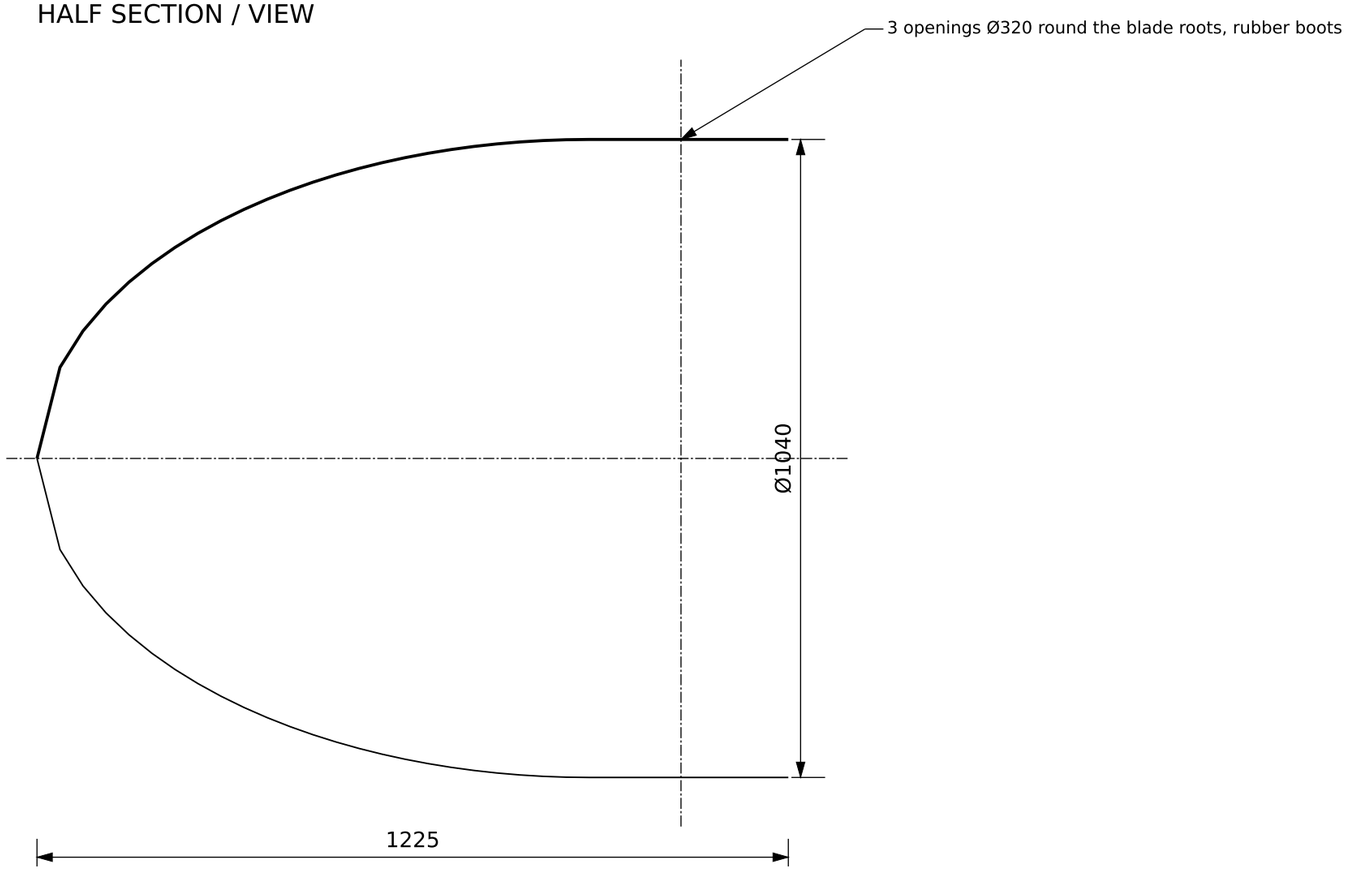
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unles eview

12 kW WIND TURBINE M8 (ICELAND) · Ø8.0 m, COLLECTIVE PITCH · 8 m TOWER

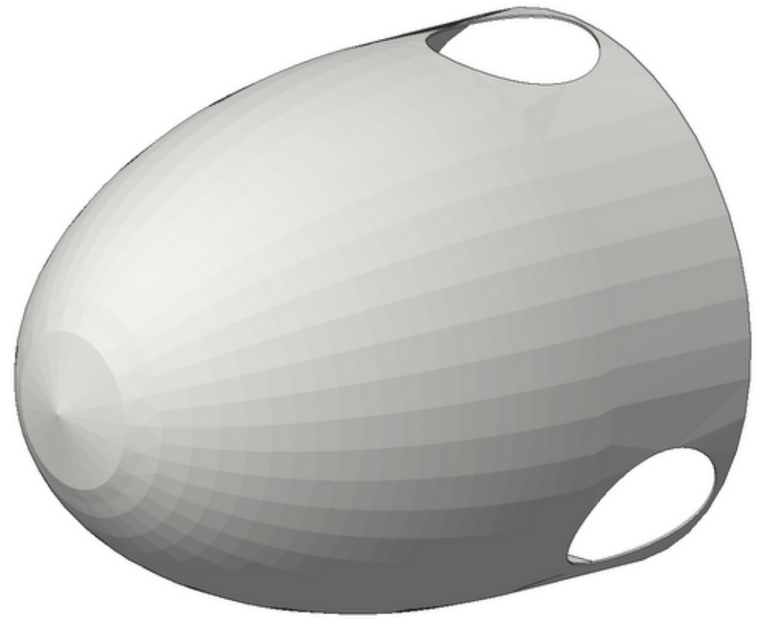
HUB WELDMENT (COLLECTIVE PITCH)

DRAWING NO.	REV / DATE	SCALE (A3)
NM8-200	A · 2026-11-06	1:8
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
S355NL, welded, stress relieved, machined	180 kg (budget)	Claude / not checked

HALF SECTION / VIEW



AXONOMETRIC VIEW (not to scale)



NOTES

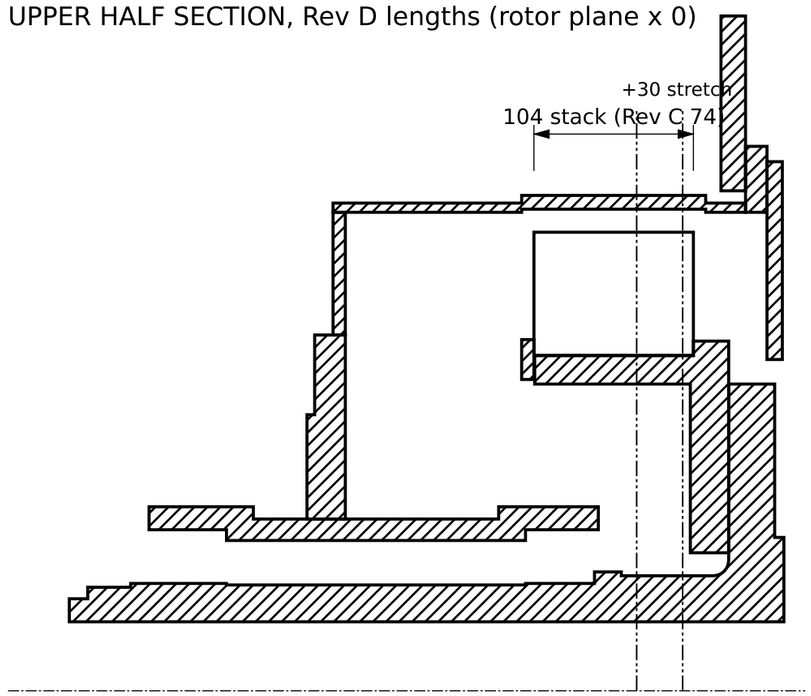
1. Ogive nose to r 520 at x −150, cylindrical to the base ring at x 175 (spindle frame, rotor plane x 0). 4 mm glass/polyester, 3 split segments with flanged joints.
2. Encloses the pitch bearings (outer edge r 485), the nose frame and the cylinder; inspection hatch Ø200 at the nose for the slip ring and the pitch electronics.
3. Fixed to the hub front ring and neck flanges with 9 brackets (A4 M8), 10 mm clearance to all moving pitch parts.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE M8 (ICELAND) · Ø8.0 m, COLLECTIVE PITCH · 8 m TOWER

SPINNER (GRP) WITH BLADE BOOTS

DRAWING NO.	REV / DATE	SCALE (A3)
NM8-210	A · 2026-11-06	1:10
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
glass/polyester 4 mm, gelcoat RAL 9010	27.6 kg	Claude / not checked

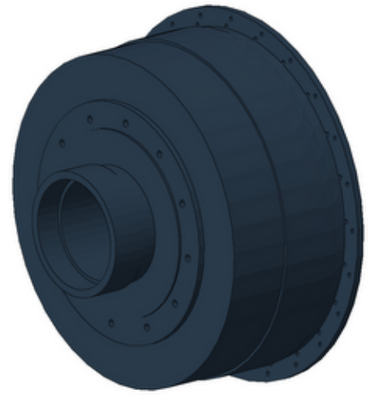


Part	Rev C	Rev D
NP1-100D spindle	x 40–476	x 40–506 (Ø150 section 400–460)
NP1-200D stator carrier	rim 74	rim 104, web unchanged
NP1-201D stator	5 rows, 74 mm	7 rows, 104 mm
NP1-203D coils	28 turns	28 turns, longer side (turn +60 mm)
NP1-210D drum	shell 239	shell 269; fins/ribs as Rev C
NP1-220F magnets	200 blocks	280 blocks (+14 spare)
Torque rated	730 Nm	991 Nm
Line EMF 100 °C	372.0 V	384.8 V
Efficiency rated	0.9334	0.9296
Drum stability margin	302	237

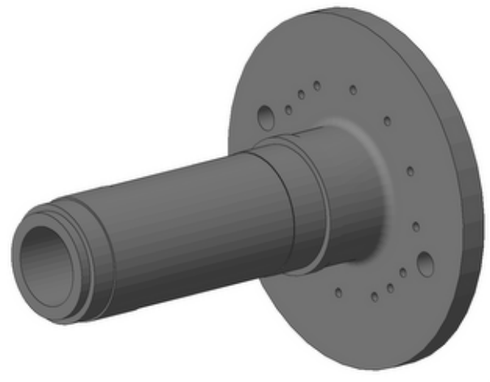
NOTES

1. All Rev C drawings stay valid with the lengths of this sheet: every feature ahead of spindle x 410 keeps its Rev C distance to the rotor plane, every feature behind x 410 moves 30 mm aft (bedplate, disc, calipers, rear cover).
2. Spindle check with Rev D lengths and M8 loads: static reserve 2.5, bearings s0 6.7 / 6.2, life factor 5.6 (spindle_m8.py).
3. Cooling as Rev C (36 inner fins, 48 outer ribs, 24 cover blades); hot spot 117 °C at 110 % (generator_m8_cooling.json). Fins and ribs are not in the 3D model.
4. Magnet comb and sleeve lengthened by 30; skew step 0.143° between the 7 rows (±0.43°).

AXONOMETRIC VIEW (not to scale)



AXONOMETRIC VIEW (not to scale)



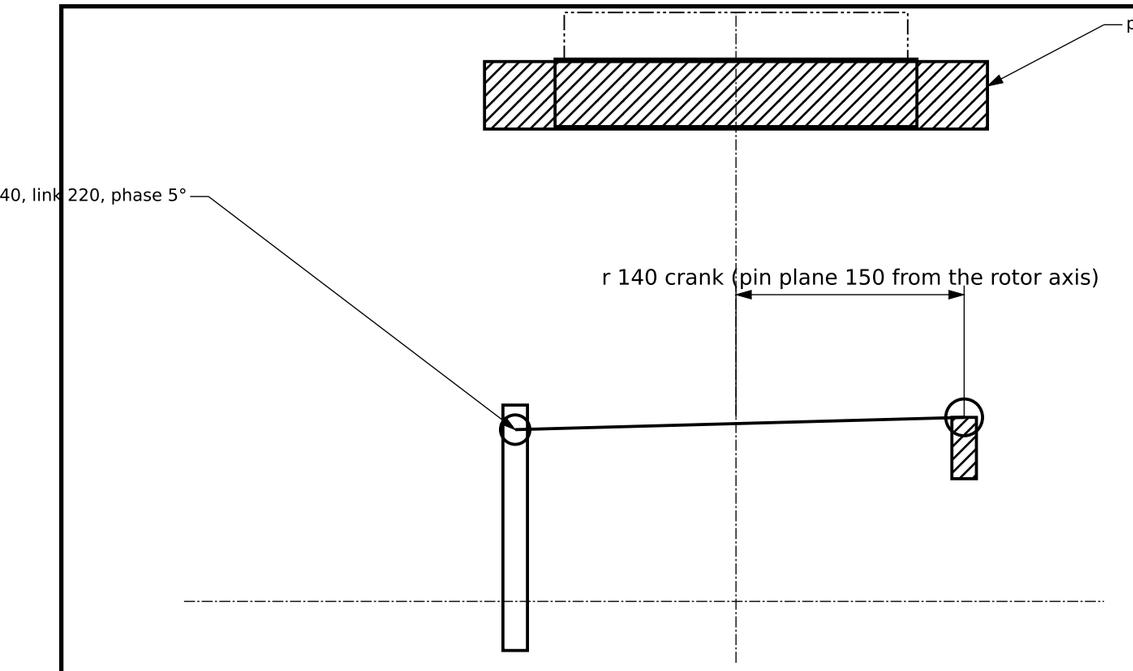
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE M8 (ICELAND) · Ø8.0 m, COLLECTIVE PITCH · 8 m TOWER

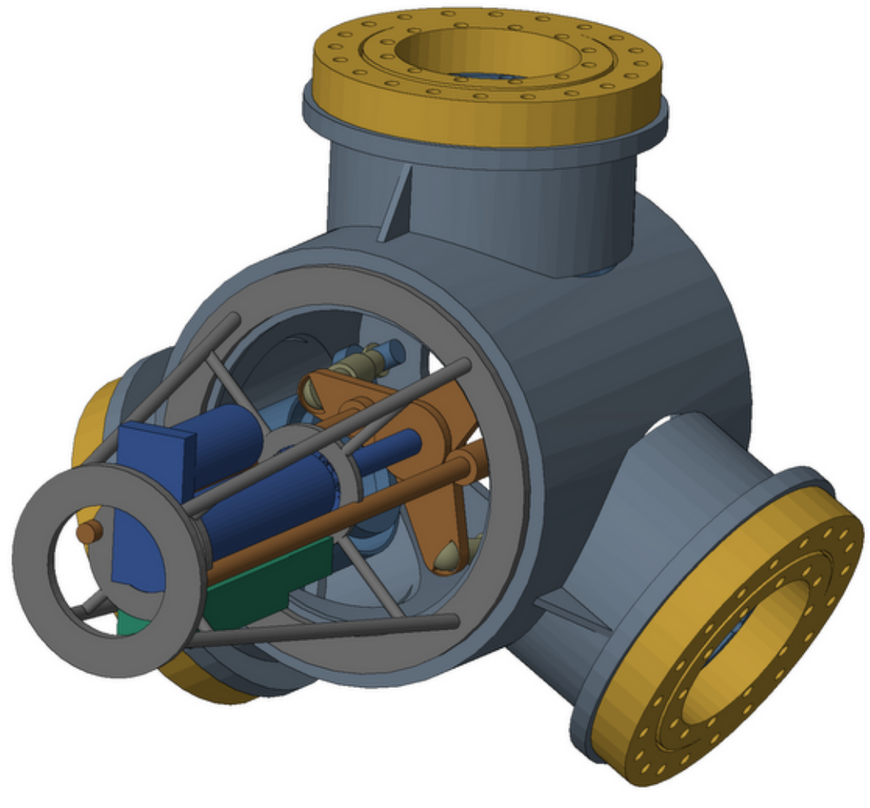
GENERATOR REV D · CHANGE SHEET (STACK 74 → 104 mm)

DRAWING NO.	REV / DATE	SCALE (A3)
NM8-215	A · 2026-11-06	1:4
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
as Rev C sheets	322 kg generator	Claude / not checked

BLADE 1 PLANE, PITCH −2° (schematic)



AXONOMETRIC VIEW (not to scale)



Pitch system (hub_yaw_m8, crank r140)	Value
Cylinder stroke	204 mm
Force at trip (design)	23.5 kN
Rated force	28.5 kN
Motor peak torque	12.6 Nm
Energy per feather stroke	4.14 kJ
Strokes from one module (EOL)	3.6
Storm hold force on screw	37.1 kN
Motor brake needed	14.2 Nm
Pitch bearing	Dpw 300, balls 30, Z 28
Bearing s0 (SLM H)	1.37
Clash check	clean at −2, 10, 30, 45, 60, 80, 92°

NOTES

1. Kinematics from analysis/hub/collective.py with R_C 140, S_P 150, R_Y 140, L 220, phase 5° (cad/hub_m8/model_m8.py). The pin plane was moved out from 118 to 150 so the crank pin clears the drum bearing housing (r ≤ 120).
2. Bearings: 4-point ungeared, 20 × M16 outer to the neck flange (PCD 360), 12 × M16 inner to the blade root (PCD 245). Grease, sealed, −40 °C grease.
3. Cylinder ≈ 28.5 kN rated, 204 mm stroke, motor brake ≥ 14.2 Nm; supercap module sized for 5.0 feather strokes at end of life; nose frame carries both.
4. Mechanical end stops at −2.5° and 93° on crank 1; limit switches at −2° and 92°.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

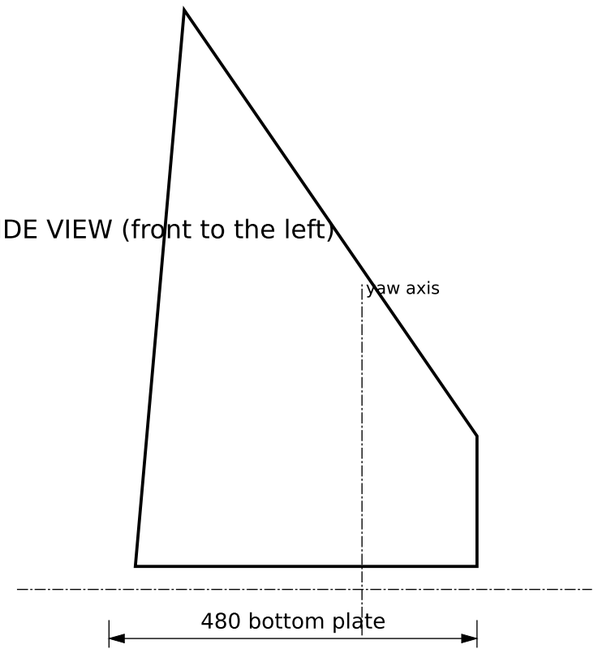
12 kW WIND TURBINE M8 (ICELAND) · Ø8.0 m, COLLECTIVE PITCH · 8 m TOWER

COLLECTIVE PITCH · BEARINGS, CRANKS, LINKS, YOKE, CYLINDER

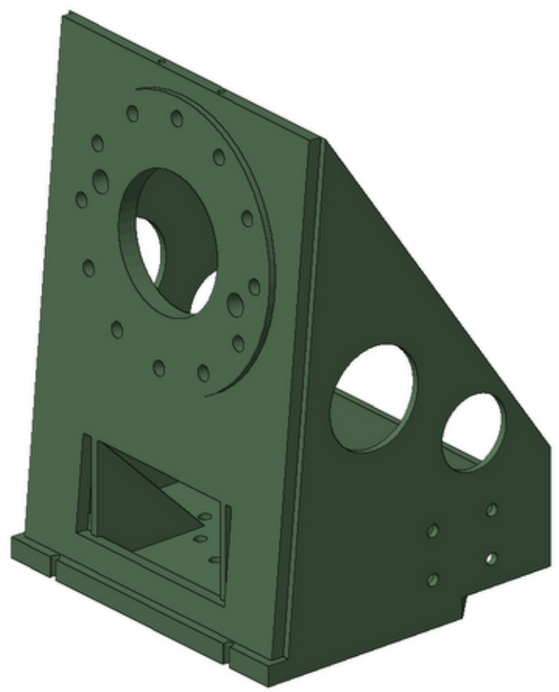
DRAWING NO.	REV / DATE	SCALE (A3)
NM8-250	A · 2026-11-06	1:5
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
see BOM	≈ 140 kg pitch parts (budget)	Claude / not checked

Plate	t mm	kg
Front plate	20	44.8
Bottom plate (yaw flange)	30	43.6
Side plates (2)	10	28.3
Check	util	
Root section	0.12	
Front plate	0.31	
Bottom plate	0.68	
Yaw bolts	0.42	
Fatigue root	0.14	

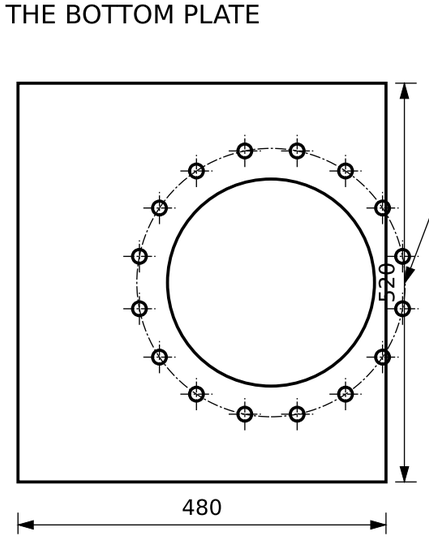
SIDE VIEW (front to the left)



AXONOMETRIC VIEW (not to scale)



PLAN OF THE BOTTOM PLATE



NOTES

1. NS6-300 geometry (doc 32) with the M8 plate thicknesses from bedplate_m8.py (decision D1): front 20, bottom 30, sides 10. Interfaces as NP1-300: spindle flange 12 × M20 PCD 350 + Ø200 spigot, pad unit and gearmotor brackets.
2. Changed for the Dpw 410 yaw ring: 16 × Ø18 on PCD 350 (was 300), central bore Ø270, pad unit bracket holes 25 mm further out. Third caliper bracket NM8-311 bolts through the front plate (4 × Ø17).
3. Stress relieve before machining; spigot bore and the yaw face machined in one set-up (parallel 0.05, square 0.05).

Dimensions mm · ISO 2768-mK · ISO 8015 · first ar

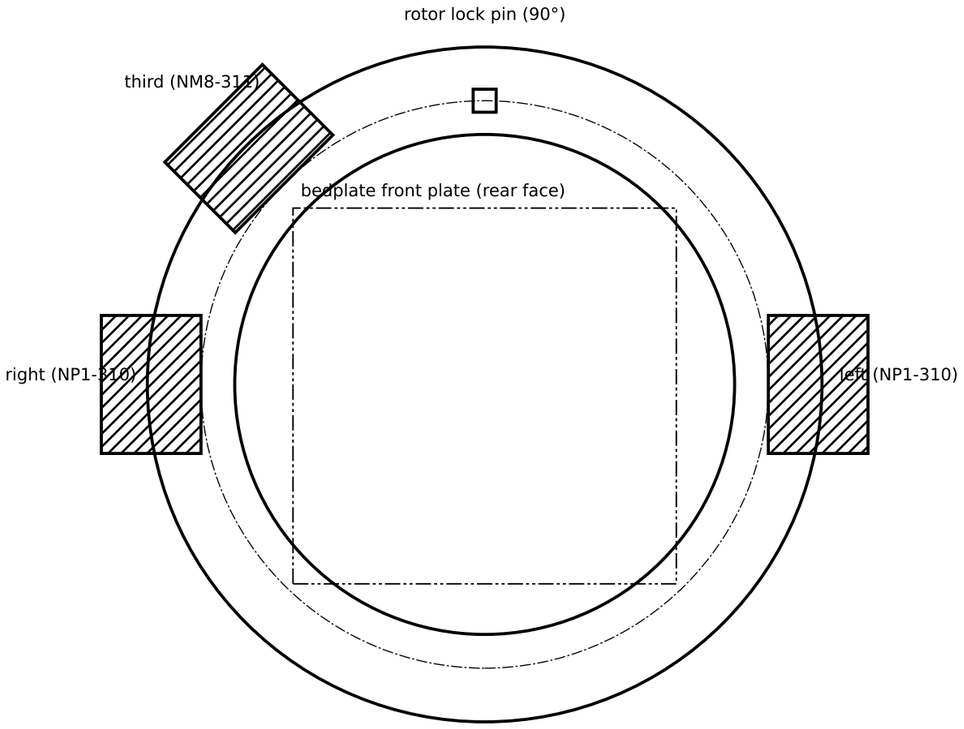
only after review

12 kW WIND TURBINE M8 (ICELAND) · Ø8.0 m, COLLECTIVE PITCH · 8 m TOWER

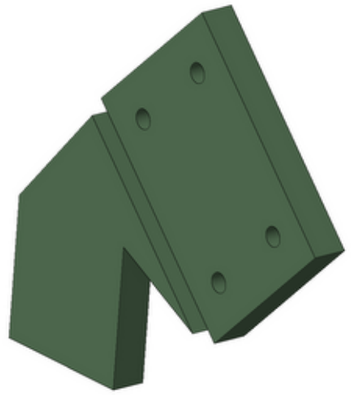
BEDPLATE WELDMENT 12 kW NP8 (LIGHT)

DRAWING NO.	REV / DATE	SCALE (A3)
NP8-300	A · 2026-11-06	1:8
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
S355J2+N plates, stress relieved, machined	136 kg (plate model), 124 kg CAD	Claude / not checked

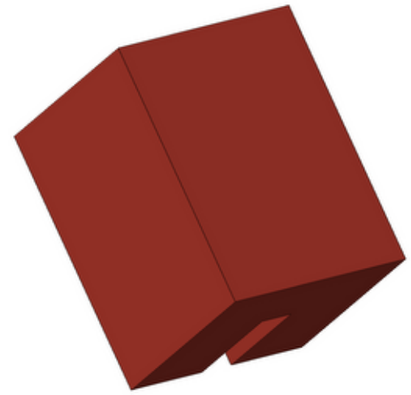
VIEW FROM THE REAR (downwind), angles from +y towards +z



AXONOMETRIC VIEW (not to scale)



AXONOMETRIC VIEW (not to scale)



Brake (spindle_m8 / hub_yaw_m8)	Value
Required holding torque	14.4 kNm (parked aero torque at 8 m, design; 18 m: 13.1)
Calipers	3 × spring-applied, 4.8 kNm each
Clamp force per caliper	16.2 kN at μ 0.40, r_eff 0.37 m
Parked aero torque (sim, char.)	9.32 kNm at 18 m, 10.59 kNm at 8 m
Disc	NP1-230 Rev B, 16 mm, unchanged

NOTES

1. The third caliper sits at 135° (upper right seen from the rear) because the rotor lock pin holder occupies 90°. It uses the same caliper and mounting plate as NP1-310 (4 × M16 axial).
2. Bracket: 25 mm mounting plate + 25 mm web + 25 mm foot, bolted 4 × M16 10.9 through the bedplate front plate outside the spindle flange bolts (r ≥ 205). Weld a5, MT 100 %.
3. Alternative (decision D2 of doc 35): two calipers with a stronger spring pack (7.2 kNm each) in the existing positions; the third bracket then goes.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

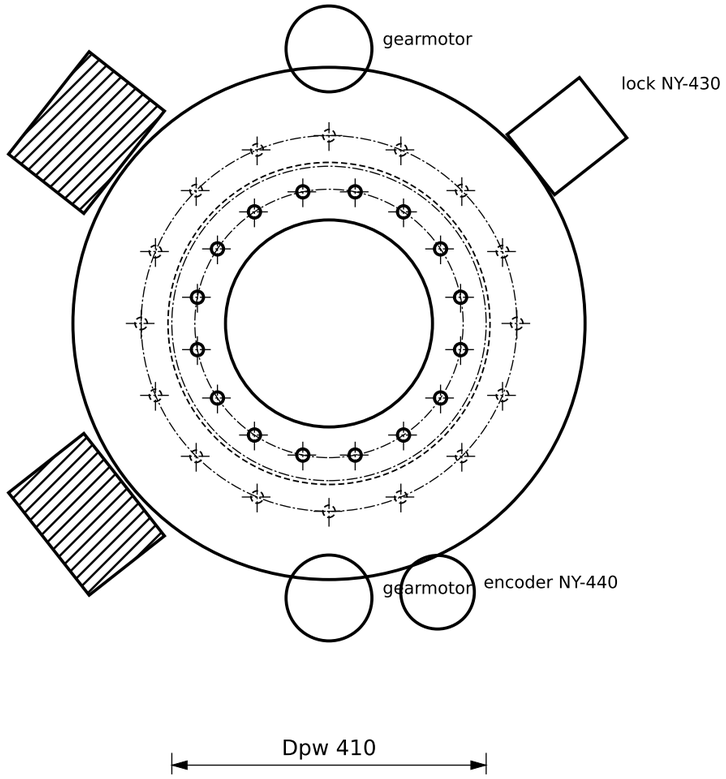
12 kW WIND TURBINE M8 (ICELAND) · Ø8.0 m, COLLECTIVE PITCH · 8 m TOWER

ROTOR BRAKE · THREE CALIPERS AND THIRD CALIPER BRACKET

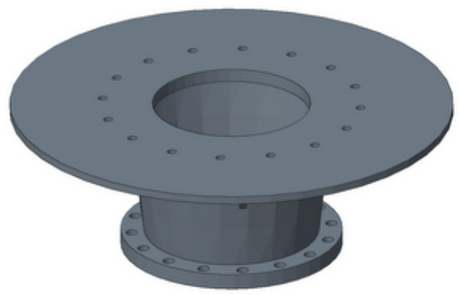
DRAWING NO.	REV / DATE	SCALE (A3)
NM8-311	A · 2026-11-06	1:8
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
bracket S355J2+N 25 plate, welded	bracket 7.0 kg	Claude / not checked

Yaw (hub_yaw_m8)	Value
Ring	Dpw 410, gear z165 m4, face 60
s0 (SLM H + shaft moment)	1.41 (Dpw 360: 0.95)
Tilt moment design	52.4 kNm
Gearbox static need	2308 Nm
Gear static S_F pinion / ring	3.39 / 2.69
Gear static S_H pinion / ring	1.54 / 1.18
Inner ring bolts	16 × M16 10.9 PCD 350 (bedplate)
Outer ring bolts	16 × M16 10.9 PCD 490 (adapter NY-400)

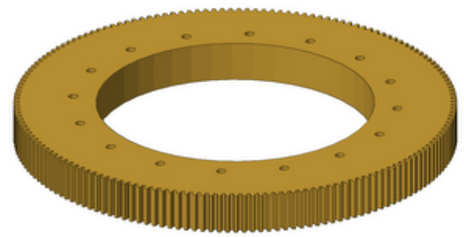
PLAN (X downwind to the right), bedplate removed



AXONOMETRIC VIEW (not to scale)



AXONOMETRIC VIEW (not to scale)



Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

NOTES

1. Doc 25 yaw system with the ring one size up (Dpw 360 fails s0 0.95 under the M8 SLM). Every radial NY dimension moves out 25 mm: adapter seat r 150–275, friction track r 275–420, pad units, lock and encoder.
2. Gearmotor centre r 358 (pinion z14 m4 on z165); NP1-420 brackets moved out 30 mm. Gearbox output ≥ 2.3 kNm static, motor brake on the input.
3. Ring bolt circles are preliminary: the supplier drawing of the chosen ring replaces this sheet's PCDs and hole sizes; bolt checks in hub_yaw_m8 used the Dpw 360 PCDs and are conservative.

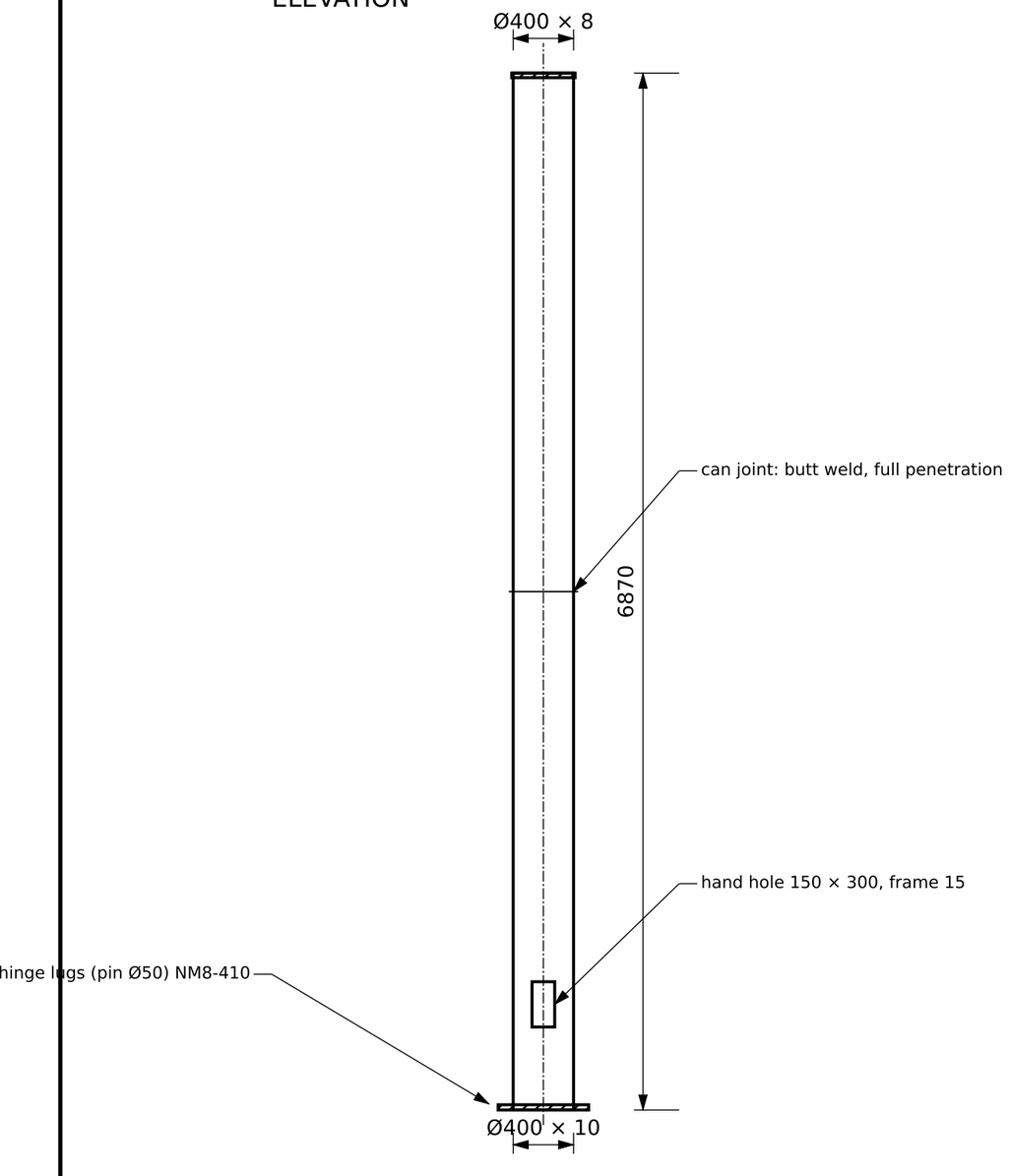
12 kW WIND TURBINE M8 (ICELAND) · Ø8.0 m, COLLECTIVE PITCH · 8 m TOWER

YAW SYSTEM LAYOUT · SLEWING RING Dpw 410

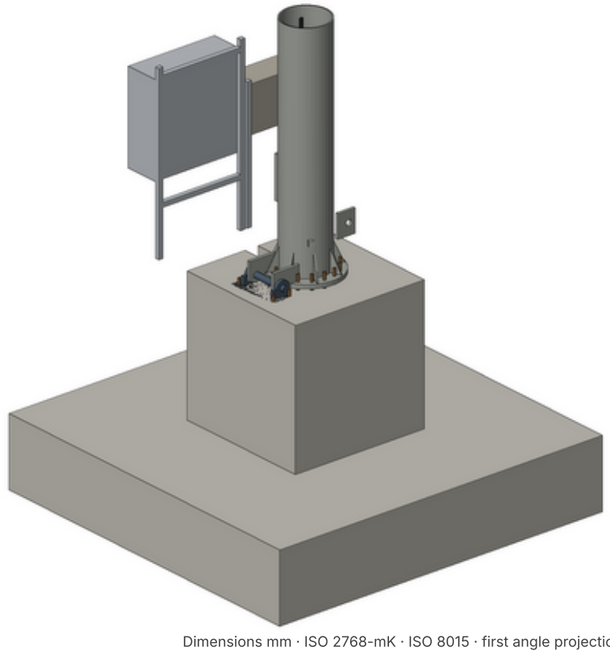
DRAWING NO.	REV / DATE	SCALE (A3)
NM8-911	A · 2026-11-06	1:8
MATERIAL	**MASS (CALC.)**	**DRAWN / CHECKED**
ring bought; NY parts per doc 25	ring ≈ 66 kg	Claude / not checked

Tower (tower8_m8, stiff_final_sim8)	Value
Mass	671 kg
f1 rigid / soft soil + ice	2.78 / 2.65 Hz
f2	22.28 Hz
3P exclusion zone	48–61 rpm
Utilisation (governing)	0.768 (earthquake 0.5 g (ground D) + operation)
Fatigue damage base / mid	0.53 / 0.07
Base flange	16 × M24 10.9 on Ø500, F_bolt 202.5 kN (F_t,Rd 254.2)
Top flange	Ø420 × 30, 16 × M20 PCD 380 to NY-400
Tilt-up base moment	150.0 kNm, line pull 66.2 kN
D/t max	50.0

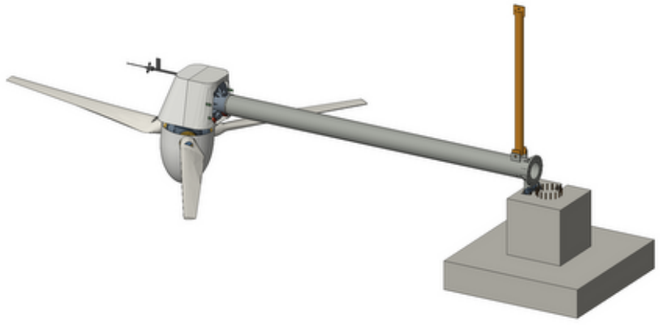
ELEVATION



AXONOMETRIC VIEW (not to scale)



AXONOMETRIC VIEW (not to scale)



NOTES

1. Stiff tower: f1 above 1.15 × 1P at 131 rpm (2.51 Hz) and below 3P/1.1 at rated (5.95 Hz). The

nferential welds full

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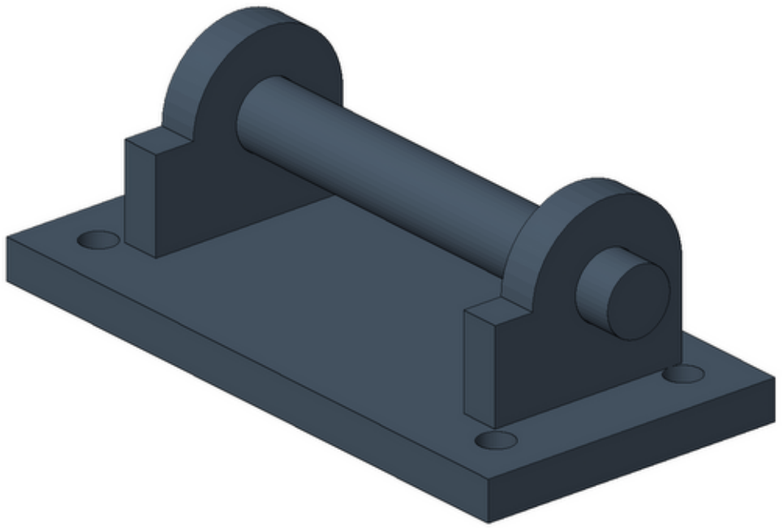
Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE M8 (ICELAND) · Ø8.0 m, COLLECTIVE PITCH · 8 m TOWER

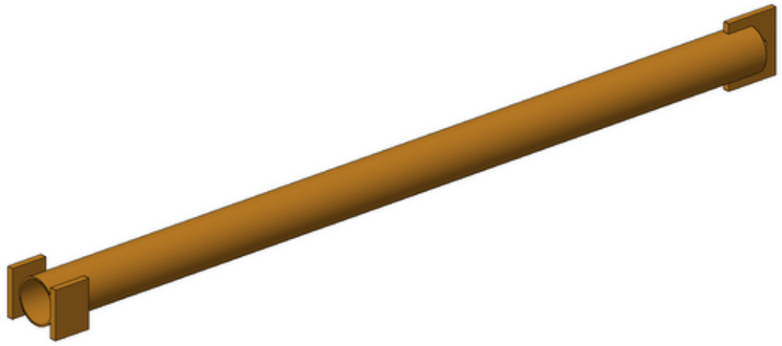
TOWER 8 m · ONE-PIECE TILT-UP MONOPOLE

DRAWING NO.	REV / DATE	SCALE (A3)
NM8-400	A · 2026-11-06	1:40
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
S355J2+N, rolled and seam welded; hot-dip galvanized	671 kg (sizing), 700 kg CAD	Claude / not checked

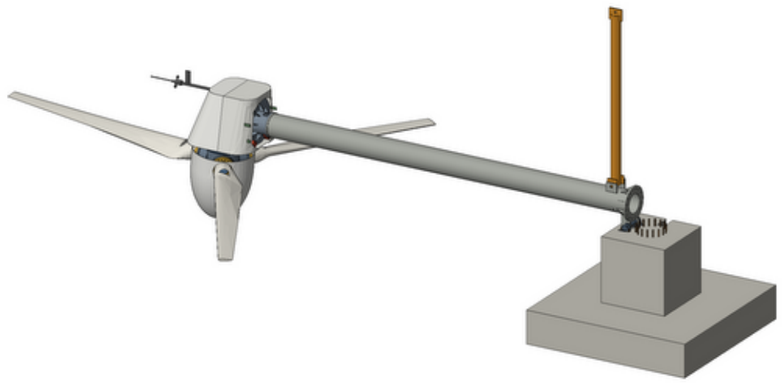
AXONOMETRIC VIEW (not to scale)



AXONOMETRIC VIEW (not to scale)



AXONOMETRIC VIEW (not to scale)



Erection (tower8_m8)	Value
Tilt-up base moment	150.0 kNm
Winch line pull (gin pole 0.33 H)	66.2 kN
Tower utilisation in tilt-up	0.37
Hinge pin	Ø50 h9, 42CrMo4+QT, double shear
Gin pole	Ø168.3 × 6.3, 3.0 m (S6 tool NS6-420)

NOTES

1. The S6 hinge and gin pole (doc 34) carry the M8 head: line pull rises with the 1.39 t head; use a 10 t winch with a 2-part line (≈ 32 kN on the drum) or a 5 t winch with a 4-part line.
2. Hinge brackets on 4 × M24 anchors in the pedestal; pin greased, secured by R-clips; tower lies on a trestle at 2.5 m during head mounting.
3. No one inside the fall zone (1.5 × tower height) while lifting; wind below 8 m/s; tag lines on two blade tips.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

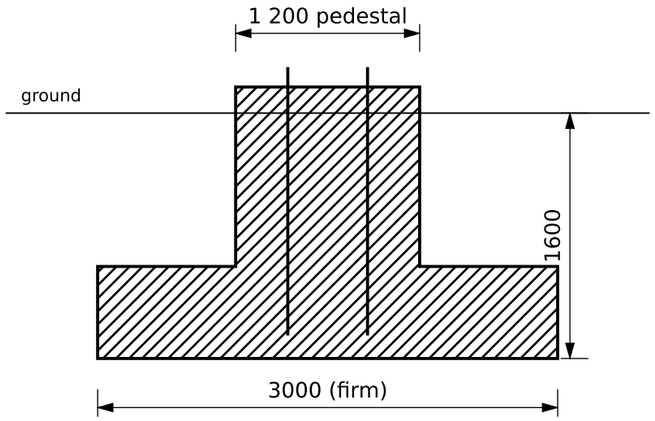
12 kW WIND TURBINE M8 (ICELAND) · Ø8.0 m, COLLECTIVE PITCH · 8 m TOWER

BASE HINGE, GIN POLE INTERFACE AND ERECTION

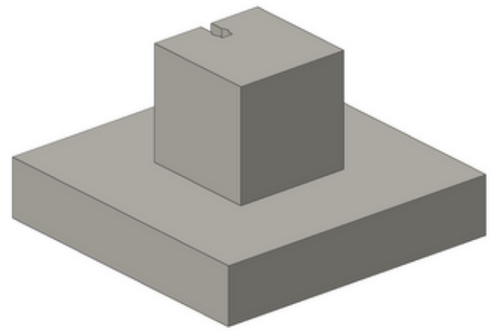
DRAWING NO.	REV / DATE	SCALE (A3)
NM8-410	A · 2026-11-06	–
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
S355J2+N 30 mm plate, galv.; pin 42CrMo4+QT Ø50 h9	37 kg	Claude / not checked

Ground class	B m	depth m	concrete m³	bearing kPa (q_Rd)
firm soil (dense gravel, moraine)	3.0	1.6	6.2	234.0 (300.0)
soft volcanic soil (loose ash, pumice sand)	3.4	1.6	7.8	97.3 (100.0)
frost-susceptible silt (thawed)	3.2	1.6	7.0	126.3 (150.0)
rock (anchored plinth)	1.6 × 1.6 × 0.8	–	2.0	8 anchors Ø20

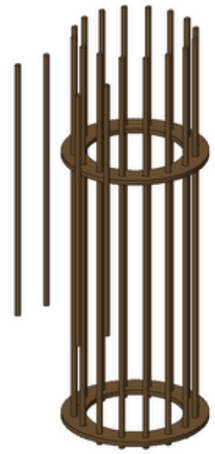
SECTION (firm ground)



AXONOMETRIC VIEW (not to scale)



AXONOMETRIC VIEW (not to scale)



NOTES

1. Anchor cage NM8-510: 16 × M24 8.8 HDG on PCD 500, two template rings, embedded 1.45 m; 4 hinge anchors M24.
2. Pedestal top 170 above ground; 30 mm grout under the base flange. Concrete C30/37 XF4 air-entrained, cover 50, rebar ≈ 75 kg/m³.
3. Ground class from a trial pit at the site (depth 2 m); frost-susceptible silt: replace 0.5 m under the slab with compacted gravel. Rock: drilled plinth with 8 anchors.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE M8 (ICELAND) · Ø8.0 m, COLLECTIVE PITCH · 8 m TOWER

FOUNDATION · SPREAD FOOTING (8 m TOWER, M8)

DRAWING NO.	REV / DATE	SCALE (A3)
NM8-500	A · 2026-11-06	1:40
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
C30/37 XF4 air-entrained; B500C rebar	6.2–7.8 m³	Claude / not checked

Joint	Fasteners	Torque / preload	Check
Blade root → pitch bearing inner ring	12 × M16 10.9 per blade	280 Nm (μ 0.12), re-torque at 50 h	F_bolt 64 kN design vs 113
Pitch bearing outer ring → hub neck	20 × M16 10.9 per blade	280 Nm	doc 35 §5
Hub → drum web	12 × M16 10.9 PCD 400	280 Nm	slip 1.7, separation 5.3
Spindle → bedplate	12 × M20 10.9 PCD 350	540 Nm	slip 6.4, separation 2.3
Stator carrier → spindle flange	12 × M12 10.9 PCD 240	110 Nm	slip 1.4
Disc → drum ring	24 × M12 10.9 PCD 662	110 Nm	slip 1.7
Caliper → bracket	4 × M16 10.9 each (3 calipers)	280 Nm	–
Third caliper bracket → bedplate	4 × M16 10.9 + nuts	280 Nm	–
Yaw inner ring → bedplate	16 × M16 10.9 HV PCD 350	280 Nm	util 0.42
Yaw outer ring → adapter	16 × M16 10.9 PCD 490	280 Nm	–
Adapter → tower top flange	16 × M20 10.9 HV PCD 380	540 Nm	ratio 0.45
Tower base flange → anchors	16 × M24 8.8 HDG, 3 nuts each	tension to 150 kN (tool), re-check at 3 months	F_bolt 202.5 kN
Hinge brackets → pedestal	4 × M24 8.8 HDG	350 Nm	–
Cover, spinner, mast, nose frame	M6–M12 A4-70	A4 table	–

NOTES

1. Torques for 10.9 zinc-flake with μ 0.12 (lubricated); mark every bolt after torquing; check 10 % of each joint at the first service (3 months) and yearly.
2. HV sets (EN 14399-4) where the joint carries the yaw moment; no washers under the nut side of HV sets other than those in the set.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE M8 (ICELAND) · Ø8.0 m, COLLECTIVE PITCH · 8 m TOWER

FASTENER AND TORQUE SCHEDULE

DRAWING NO.	REV / DATE	SCALE (A3)
NM8-900	A · 2026-11-06	–
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
ISO 4014/4017, EN 14399-4 (HV), 10.9 zinc-flake unless stated	–	Claude / not checked

Part	Specification	Interface	Qty
Pitch bearing (NM8-250)	4-point, Dpw 300, balls 30, Z 28, s0 ≥ 1.25 under SLM H	outer 20 holes PCD 360, inner 12 × M16 PCD 245	3
Pitch cylinder + motor (NM8-270)	≥ 28.5 kN rated, 37 kN static, 204 mm stroke, motor brake ≥ 14.2 Nm	trunnion in the nose frame	1
Supercap module (NM8-280)	≥ 4.1 kJ per stroke × 3.6 at EOL, −30 °C heater	nose frame tray	1
Main bearings (NM8-913)	32028 X tapered roller, C 280 kN, C0 430 kN	Ø140 j6 / f6, Ø210 P7	2
Brake calipers (NM8-915)	spring-applied, 4.8 kNm each on the Ø880 × 16 disc, ≥ 16.2 kN clamp	4 × M16 axial	3
Yaw slewing ring (NM8-911)	Dpw 410, external gear z165 m4, face 60, s0 ≥ 1.25	PCD 350 / 490, 16 × M16 each	1
Yaw gearmotor (NM8-912)	≥ 2.3 kNm static output, pinion z14 m4, motor brake	NP1-420 bracket	2
Magnets (NP1-220F)	N45SH 30 × 15 × 12, NiCuNi, 280 + spares	comb + sleeve	280
Slip ring	6 power + 6 signal, IP65, −30 °C	spindle nose Ø120	1
Overspeed sensors	M18 inductive PNP, 2 channels	NP1-321 seats	2
Converter + controller	12 kW, doc 12 / doc 35 §8	base cabinet	1

NOTES

1. Replace the 3D envelopes with the supplier models before release and rerun cad/turbine_m8/clash8.py.
2. Low-temperature grades (−30 °C operation, −40 °C storage) for all lubricants, seals and electronics.

Dimensions mm · ISO 2768-mK · ISO 8015 · first angle projection · edges broken 0.5 unless stated · NOT CHECKED: prototype release only after review

12 kW WIND TURBINE M8 (ICELAND) · Ø8.0 m, COLLECTIVE PITCH · 8 m TOWER

BOUGHT PARTS SPECIFICATION (ENVELOPES AND INTERFACES)

DRAWING NO.	REV / DATE	SCALE (A3)
NM8-910	A · 2026-11-06	–
MATERIAL	MASS (CALC.)	DRAWN / CHECKED
see lines	see lines	Claude / not checked